MINIMALLY SUFFICIENT EXPERIMENTAL DESIGN USING MATHEMATICAL MODELING



ODALYS & LAUREN



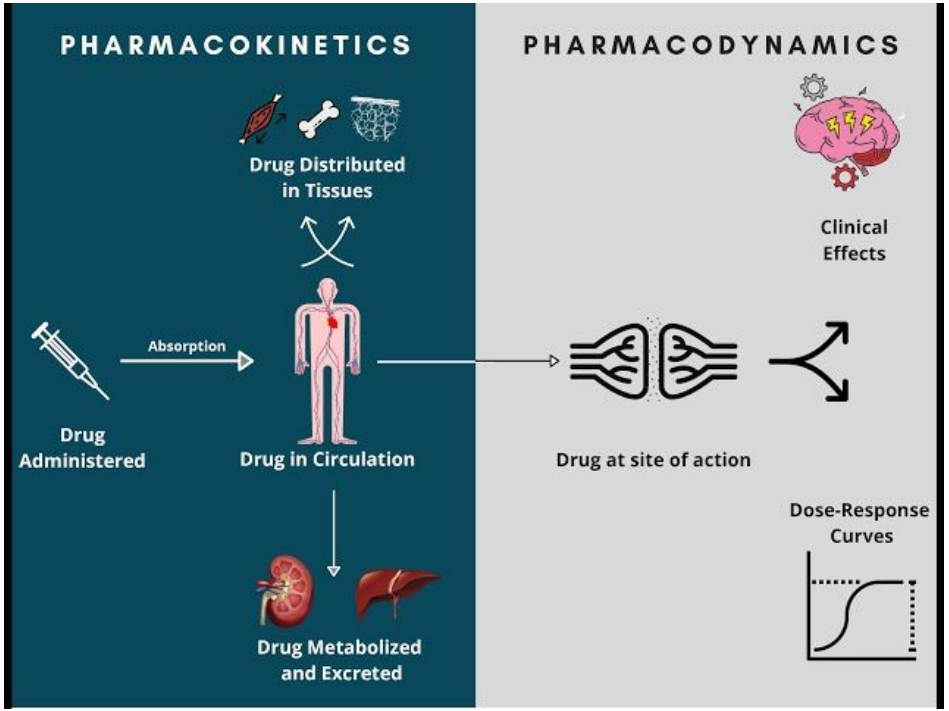
Pharmacokinetics & Pharmacodynamics

Pharmacokinetics (PK):

- How drug moves throughout body
- What does the body do to the drug?
  - Absorption, Distribution, Metabolism, and Elimination

Pharmacodynamics (PD):

- Relationship between drug concentrations at site of action and resulting effect
- What does the drug do to the body?
  - Reduce the size of a tumor



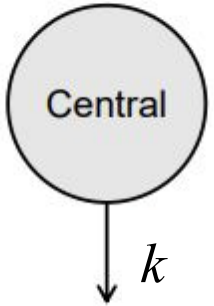
PK Compartment Modeling: 1 Compartment



Equation:

$$\frac{dc}{dt} = -kc$$

Diagram:



Initial Condition: $c(0) = c_0$

- How much drug enters the central compartment at an initial time point

PK Compartment Modeling: 2 Compartment

Equations:

- Central Compartment

$$\frac{dc_1}{dt} = jk_{21}c_2 - k_{12}c_1 - kc_1$$

- Peripheral Compartment

$$\frac{dc_2}{dt} = dk_{12}c_1 - k_{21}c_2 - kc_2$$

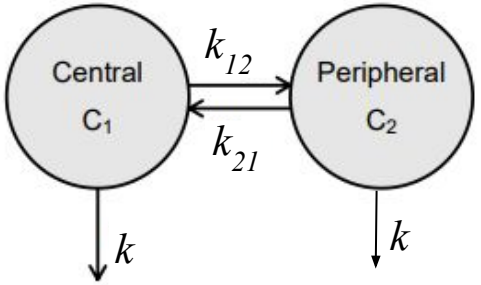
$j = V_2/V_1$

$d = V_1/V_2$

V_1 :volume from central compartment

V_2 :volume from peripheral compartment

Diagram:



$$c_1(0) = \alpha$$

Initial Condition: $c_2(0) = 0$

- Inject into central compartment, initially no drug concentration in the peripheral compartment

Receptor Binding (PD) Model

- A drug-receptor complex forms when a drug molecule combines with a receptor to produce a new product, the drug-receptor complex. It is the formation of this complex that triggers drug action.

$$\frac{dR}{dt} = -k_{onA}C_A R + k_{offA}R_{CA}\text{: Unbounded Receptor}$$

$$\frac{dR_{CA}}{dt} = k_{onA}C_A R - k_{offA}R_{CA}\text{: Bounded Receptor}$$

Example:

$$C_A + R \underset{k_{offA}}{\overset{k_{onA}}{\rightleftharpoons}} RC_A$$

C_A : drug molecule from drug A

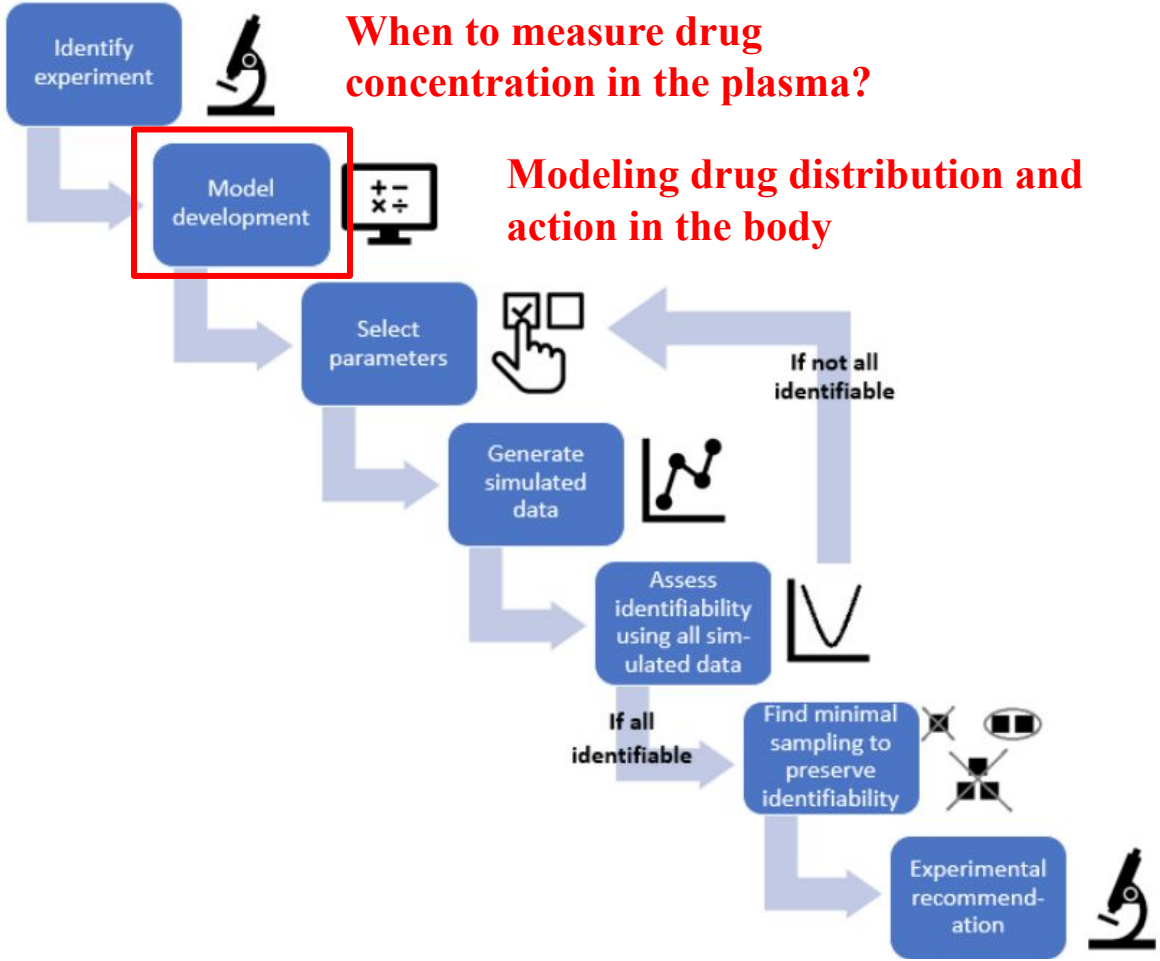
R : receptor

RC_A: drug-receptor complex

k_{onA} : binding rate

k_{offA} : dissociation rate

Experimental Design



Project Model

$$\frac{dD_p}{dt} = dk_2D_c - k_{on}D_pR + k_{off}DR - k_3D_p - k_1D_p\text{: Peripheral}$$

$$\frac{dR}{dt} = -k_{on}D_pR + k_{off}DR\text{: Unbounded Receptor}$$

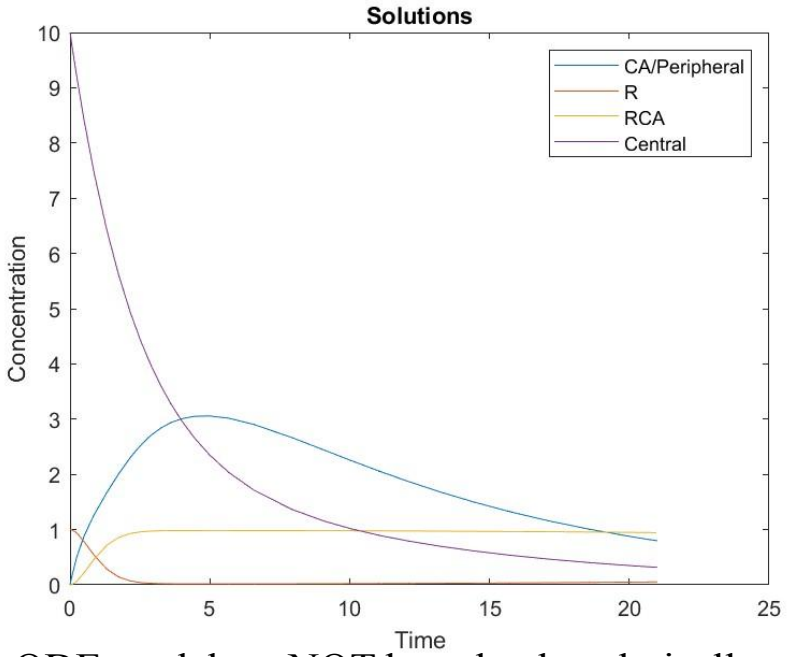
$$\frac{dDR}{dt} = k_{on}D_pR - k_{off}DR\text{: Bounded Receptor}$$

$$\frac{dD_c}{dt} = -k_1D_c - k_2D_c + jk_3D_p\text{: Central}$$

Parameter	Type of Rate
k_{on}	binding
k_{off}	disassociating
k_1	distribution
k_2	distribution
k_3	distribution
j	volume ratio
d	volume ratio

Parameter	Value
k_{on}	1
k_{off}	0.05
k_1	0.1
k_2	0.25
k_3	0.1
j	1
d	1

Initial Conditions = [0 1 0 10]



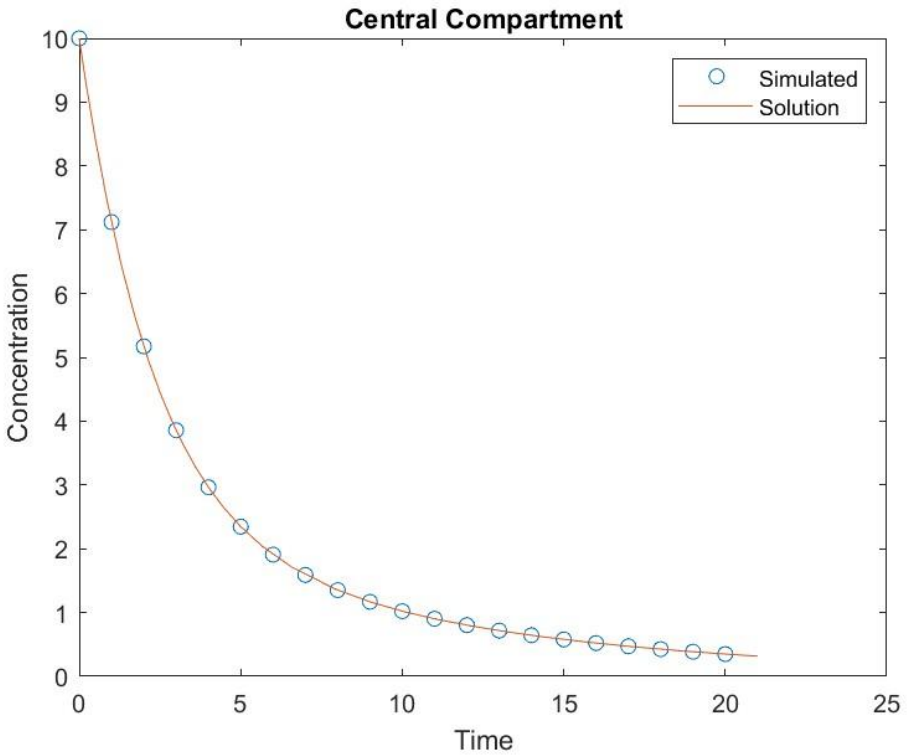
ODE model canNOT be solved analytically. Instead, it is solved in MATLAB using the built-in ODE45 solver.

Goal:

What is the minimum number of experimental time points needed to confidently determine parameters in an ODE model?

Extracting Simulated Data from Central Compartment

- Rather than use noisy, real data to answer this question, we will instead use **simulated data.**
- This simulated data is extracted directly from the model, and thus assumes the model is the "ground truth" for drug PK/PD.
- Since the data that would actually be measured is drug concentration of the central compartment, we extract this data from the model solution to use as our simulated data.
- The blue points represent extracted concentration (simulated data) at that specific time using the original parameter values.



Fitting Model to Simulated Data

- The cost function we minimize is the sum of the square errors (SSE)

$$\text{SSE} = \sum_{t=1}^{20} (y_{model}(t) - \text{concentration}(t))^2$$

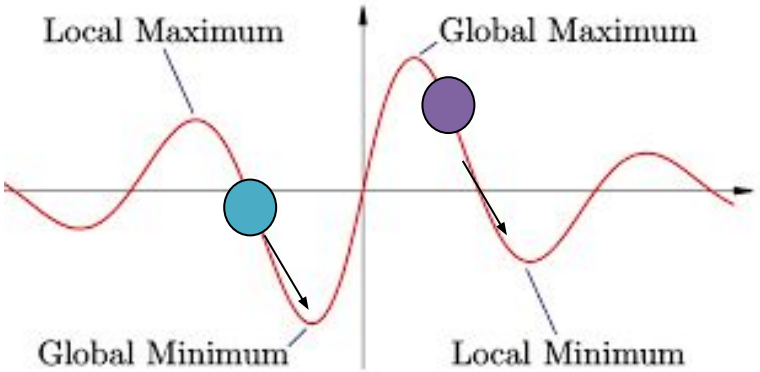
where $y_{model}(t)$ is the model-predicted value of drug concentration in the central compartment, and concentration(t) is the value of drug concentration in the simulated data.

- To find the parametrization of the model that minimizes SSE, we used MatLab's built-in function *fmincon*.
  - *fmincon* is a gradient-based method that is designed to work on problems where the cost function and constraint are both continuous and have continuous first derivatives.
  - We impose *constraints* of the value of each parameters based on biological considerations about what each parameter represents.

Parameter	LB	UB
k_{on}	0	3
k_{off}	0	0.15
k_1	0	0.3
k_2	0	0.75
k_3	0	0.3
j	0	3
d	0	3

Fitting Model to Simulated Data

- Thus our goal is to find the set of 7 model parameters that best describe the simulated data.
  - We note that we are fitting the model to perfect data that was actually *generated from the model*. Thus, it is theoretically possible that the algorithm could identify the perfectly correct model parametrization with a SSE of 0.
  - However, this is a high-dimensional optimization problem, and most optimization algorithms tend to converge to local versus global minimum.
- Since we want to reduce the cost function as much as possible, we developed a multistart algorithm. Multistart uses random starting points to improve the chances of obtaining the global minimum. Without multistart, certain starting points can keep you stuck near a local minimum.

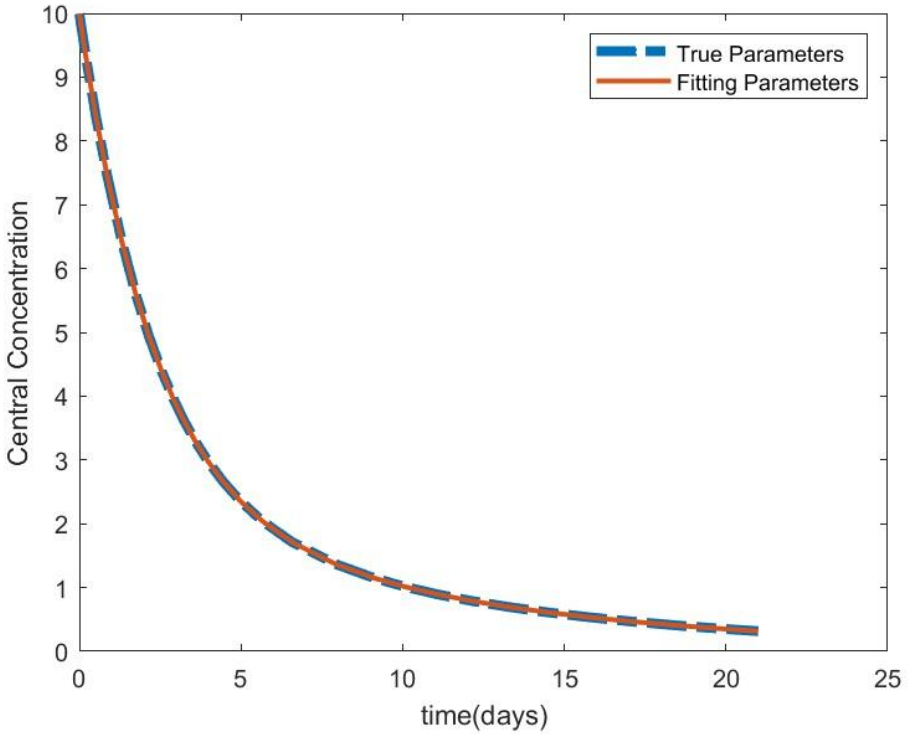


Fits to Trained Data (Central Compartment)

Parameter	Fit Value	True Value
k_{on}	1.0015	1
k_{off}	0.0509	0.05
k_1	0.1127	0.1
k_2	0.2373	0.25
k_3	0.0874	0.1
j	1.1463	1
d	1.0519	1

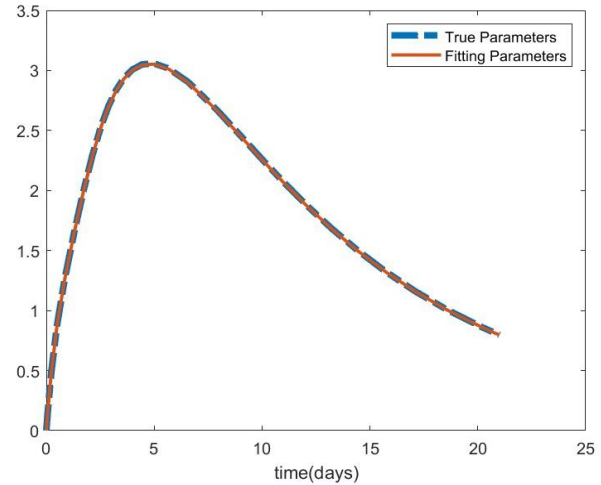
SSE: 7.6021e-10

- While very small, note that the optimization algorithm is not finding the perfect parameterization, for which each fit value would be the same as the true value, and the SSE would be 0.

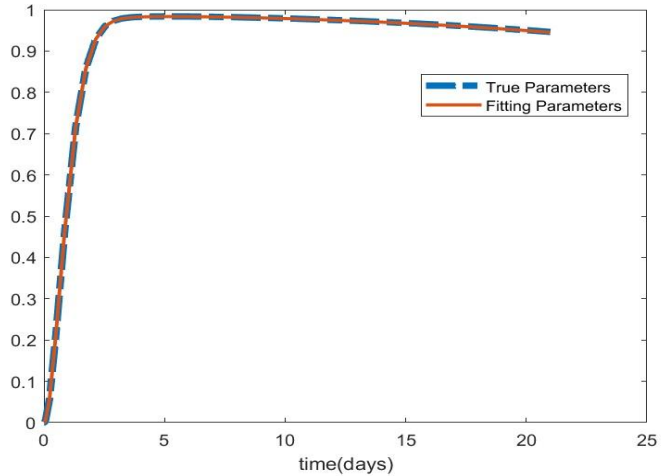


Fits to Other Compartments (Data Not Trained On)

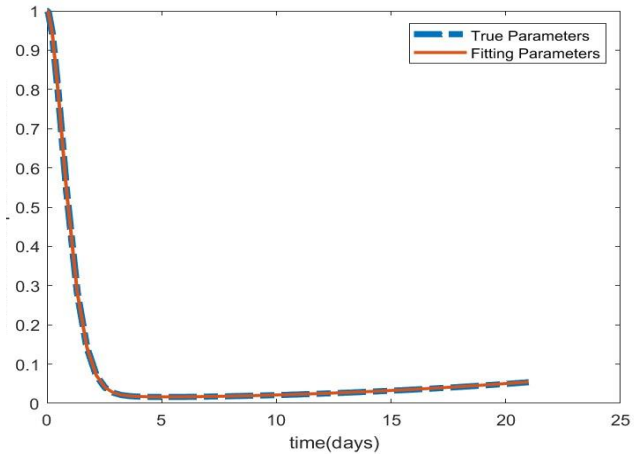
Peripheral Concentration (D_p)



Drug-Receptor Complex (DR)



Unbound Receptor (R)



Note excellent fit to all untrained data!

Profile Likelihood Method:

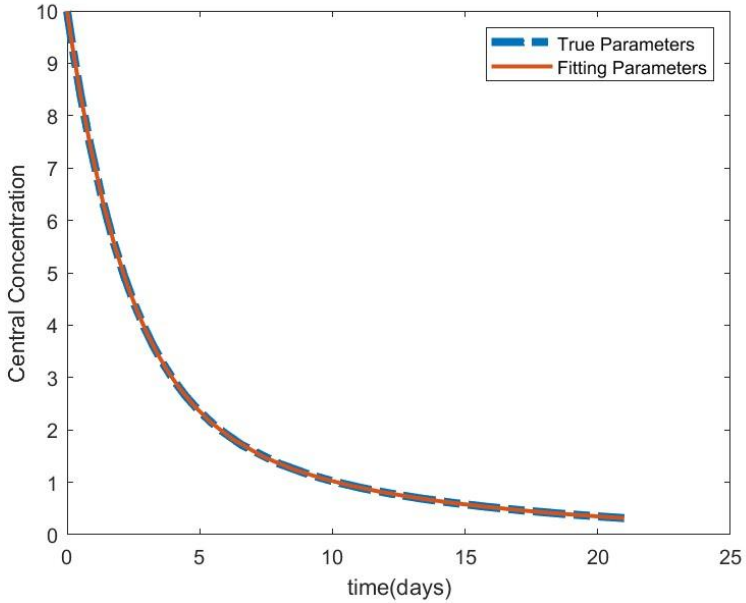
Which parameters can the model identify with confidence, given available data?

Can Other Parameterizations Describe Data Nearly Equally Well?

Parameter	Other	Fit Value	True Value
k_{on}	0.9405	1.0015	1
k_{off}	0.0622	0.0509	0.05
k_1	0.1667	0.1127	0.1
k_2	0.1836	0.2373	0.25
k_3	0.0343	0.0874	0.1
j	2.9493	1.1463	1
d	1.3528	1.0519	1

SSE: 1.9794e-07

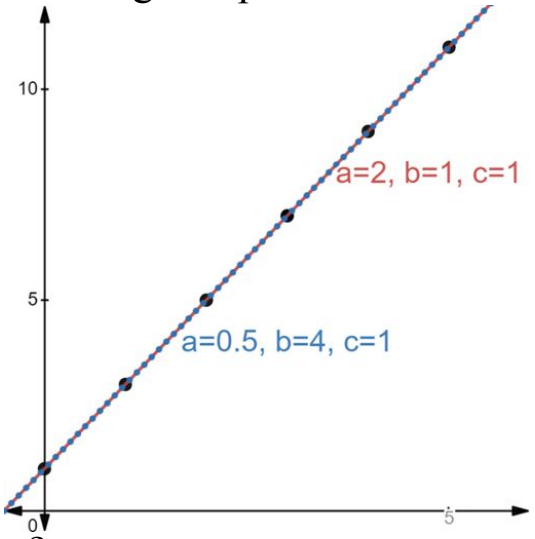
- Fit is still excellent, though the parametrization is quite different from the fit parameterization, and even more distinct from true parameterization



Raises the question of how much we can *trust* the model parametrization

Quantifying Confidence in Model Parameterization

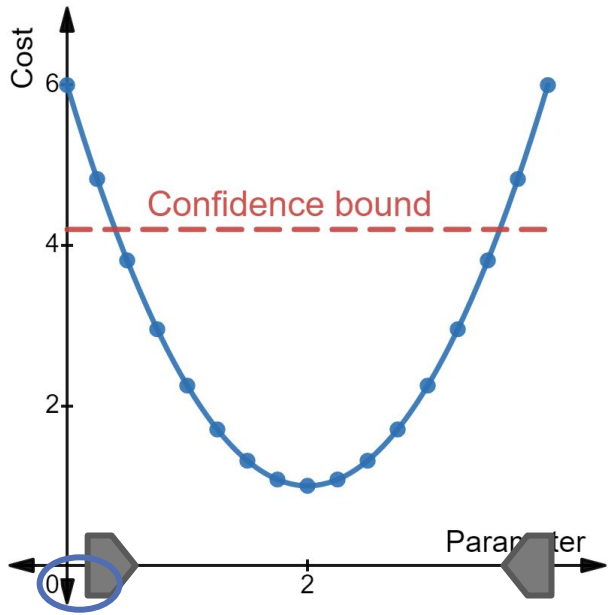
- Motivating question: is it possible to uniquely determine model parameters from data?
- **Structural identifiability**: Can parameters be uniquely determined given perfect data?
  - *y_model=abt+c* model is not structurally identifiable for perfect data given here: no unique parameterization exists
  - Can fix by re-parameterizing model *m=ab*: *y_model=2t+1*
- Necessary condition for practical identifiability
- **Practical identifiability**: What if we have real and noisy data?



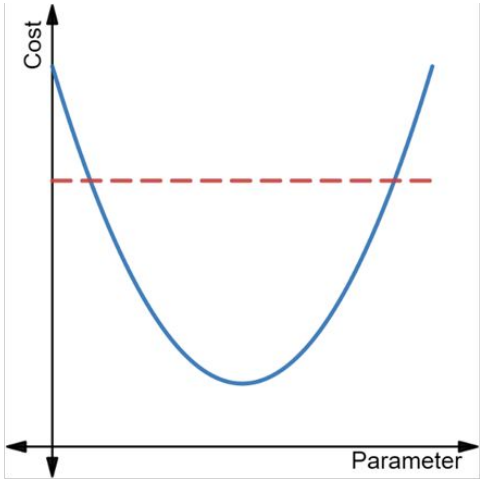
Assessing Practical Identifiability: Profile Likelihood

Profile likelihood method for parameter $\boldsymbol{\theta}$

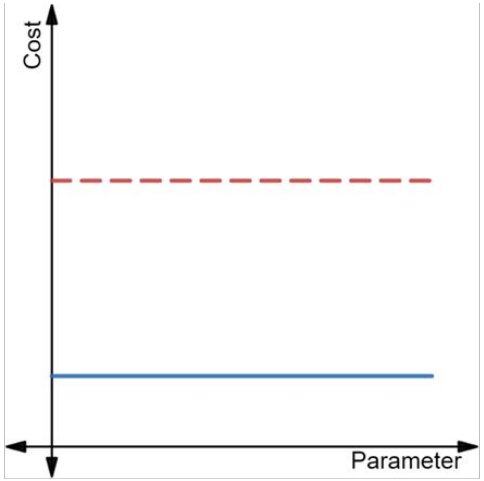
1. Determine a range for the values of $\boldsymbol{\theta}$
2. Fix $\boldsymbol{\theta}=\boldsymbol{\theta}^*$ where $\boldsymbol{\theta}^*$ is a value in the range
3. For the fixed value in Step 2, find the parameter set that minimizes the cost function
4. Save the optimal value of the cost function for $\boldsymbol{\theta}= \boldsymbol{\theta}^*$
5. Repeat the process in Steps 2 – 4 for a discrete set of values in the range of the parameter $\boldsymbol{\theta}$



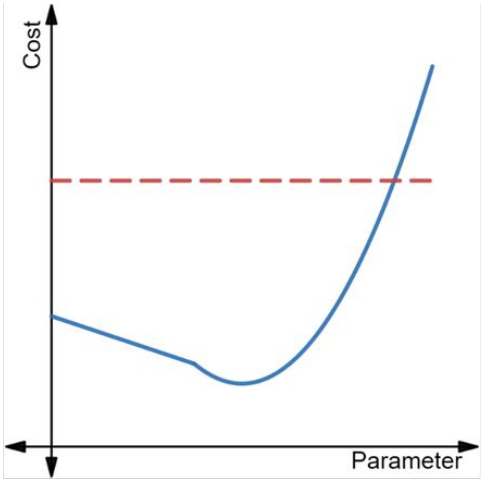
Interpreting Profile Likelihood Curves



Practically identifiable: profile likelihood has parabolic shape with clear minimum at optimal parameter set

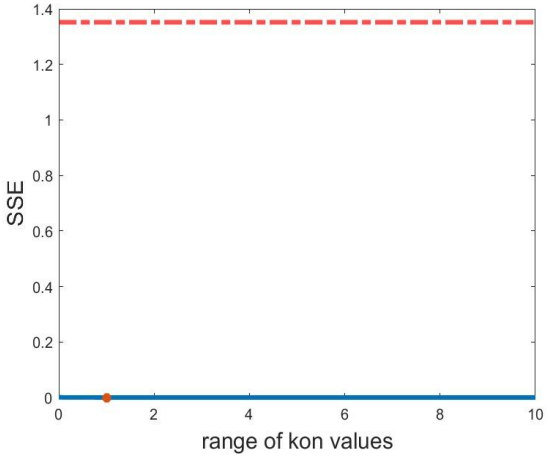


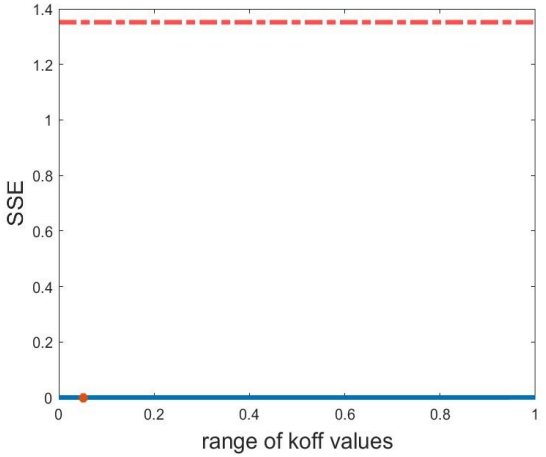
Structurally non-identifiable: profile likelihood is flat, meaning an infinite set of parameters give equally good fits to data

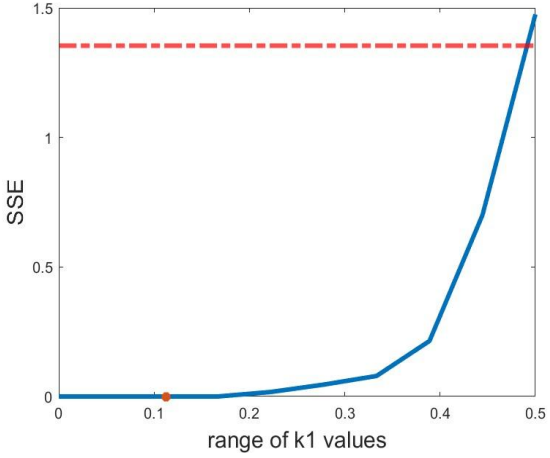


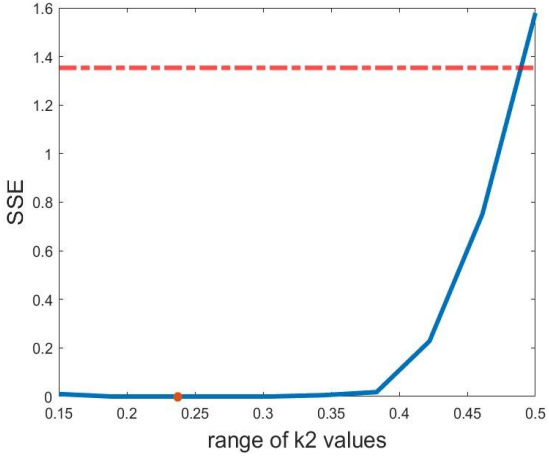
Practically non-identifiable: structurally identifiable model, but optimal parameter set cannot be clearly determined from data

Identifiability of our Model Parameters



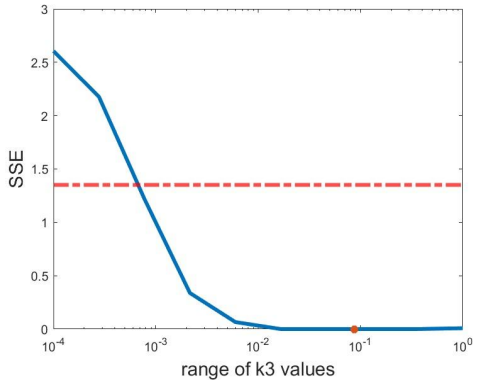


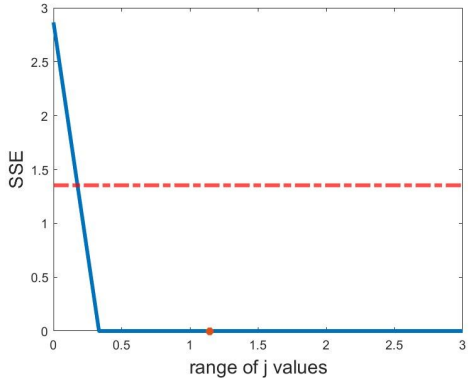


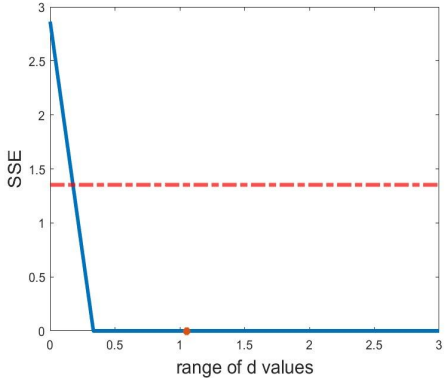


Narrowing Down to a Subset of Model Parameters

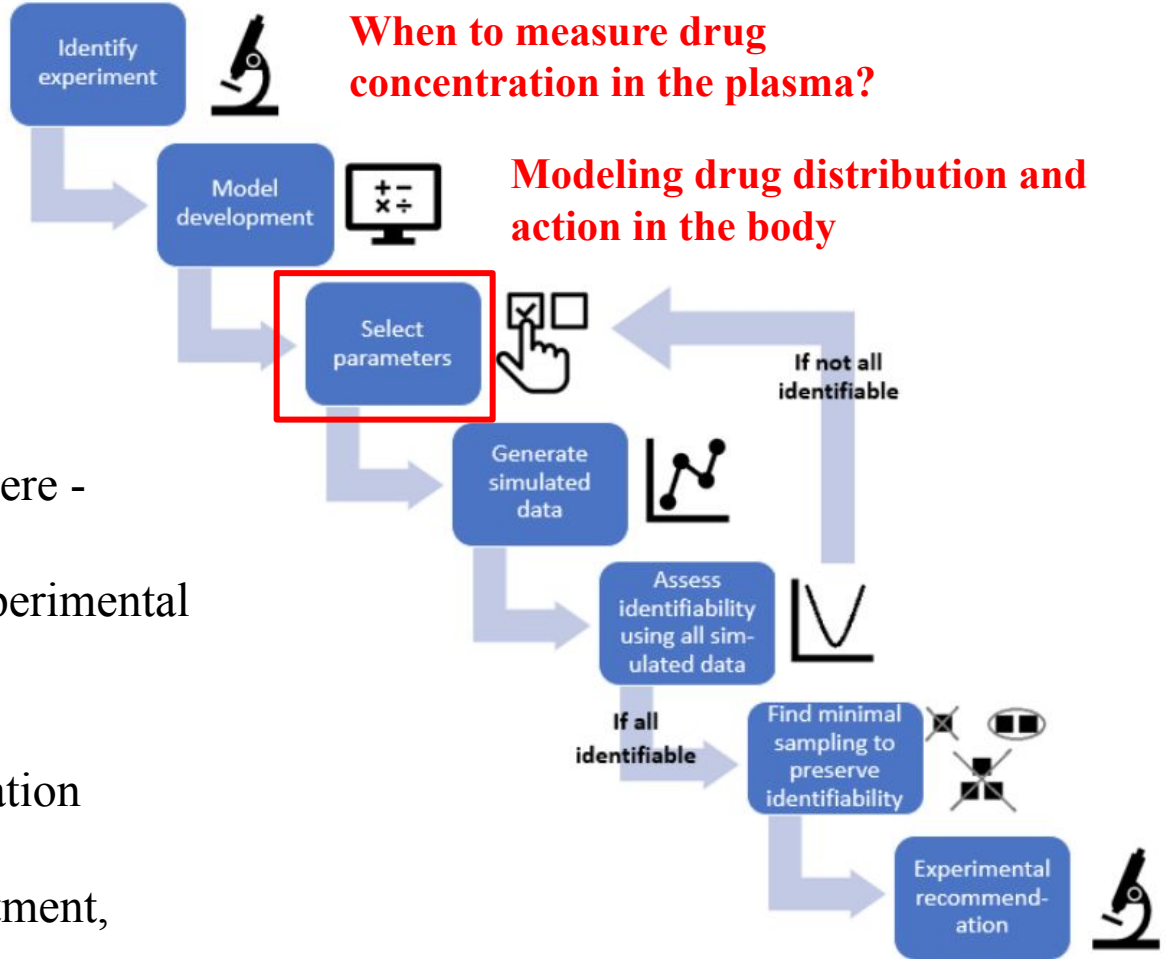
- Even the “ideal” scenario of collecting drug concentration in the central compartment daily does not give us enough data to confidently parameterize the model (no parabolic profiles)
- Without identifiable parameters, we cannot trust model predictions: as different model parameterizations that well-describe the data could make very different predictions!







Experimental Design

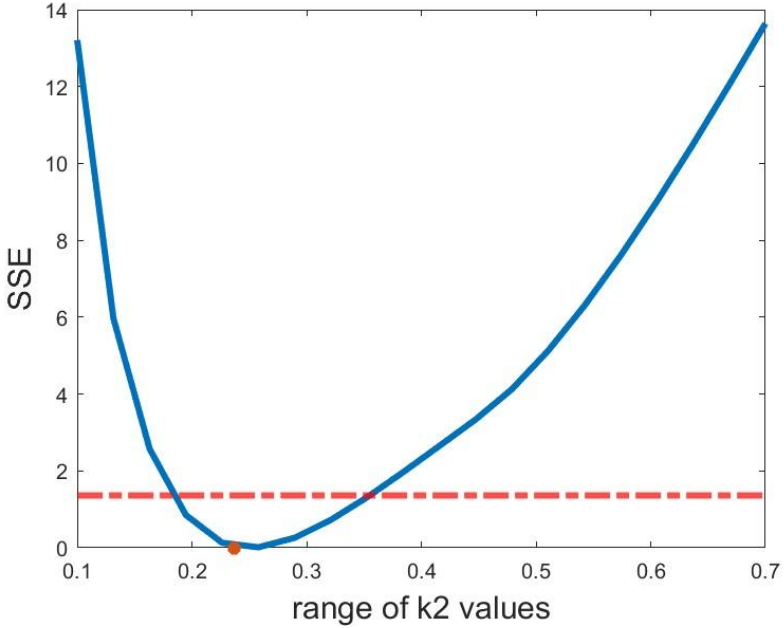


We turn to biology to help here - some parameters can be well-approximated from experimental data. We will fix these:

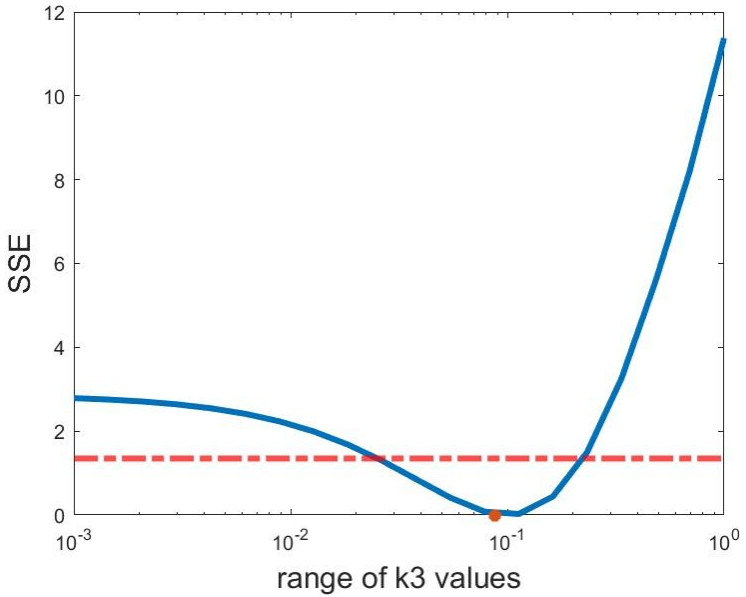
- Decay rate of drug, k_1
- Association and dissociation rates, k_{on} and k_{off}
- Volume of each compartment, giving us j and d

Subset of Parameter Profiles Using All Data Points (Fixing All Non-Profiled Parameters)

k_2 Profile

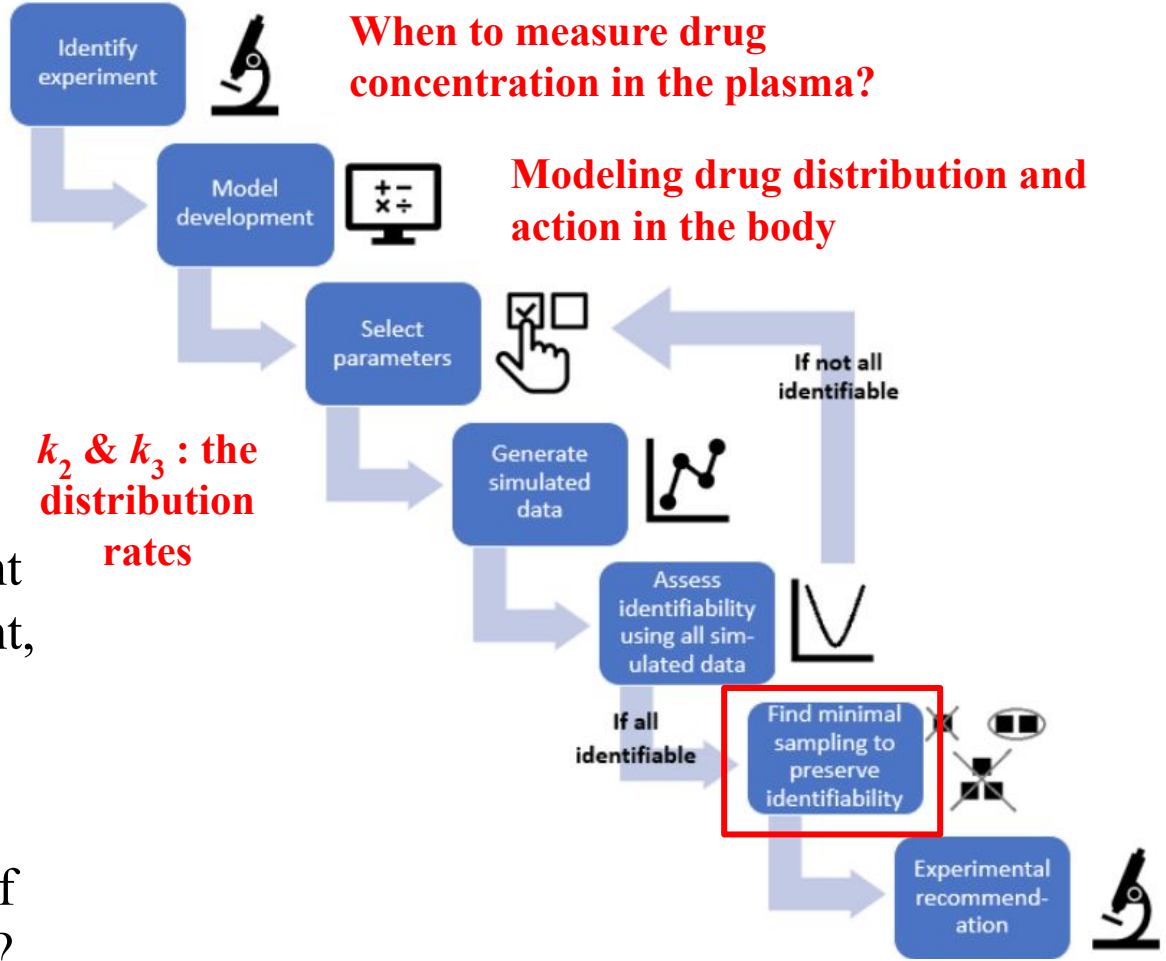


k_3 Profile



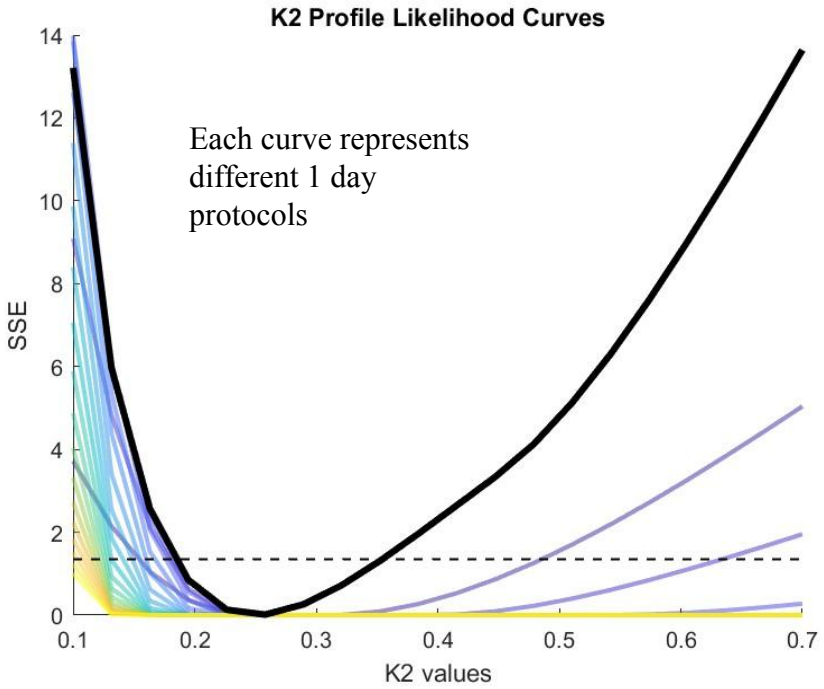
k_2 and k_3 are distribution rates; If we collect data daily, and we know the value of all other parameters, k_2 and k_3 are practically identifiable!

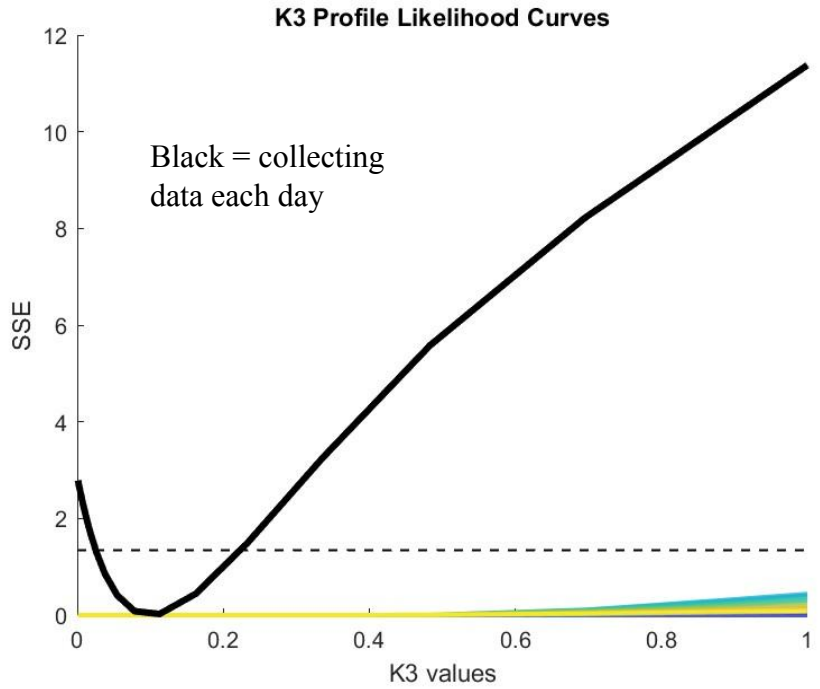
Experimental Design



Back to Goal: If we want to replicate the experiment, what is the minimal number of days of data collection needed to preserve identifiability of the subset of parameters?

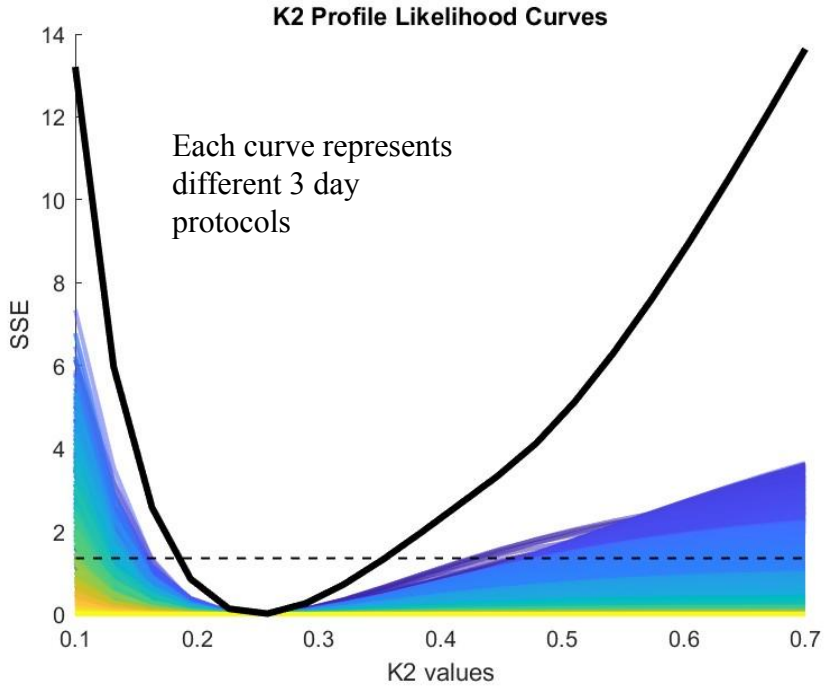
Non-Optimal Experimental Designs: 1 Data Point

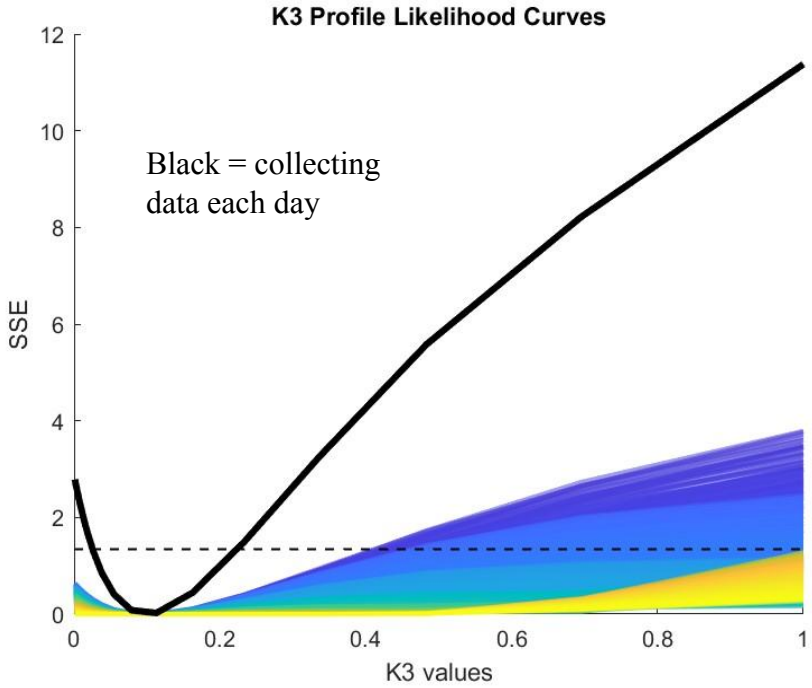




We want a parabolic shape in the curve, as that indicates practical identifiability. k_2 does have two points which cross the threshold. k_3 does not meet the threshold value, which indicates we need to collect more data.

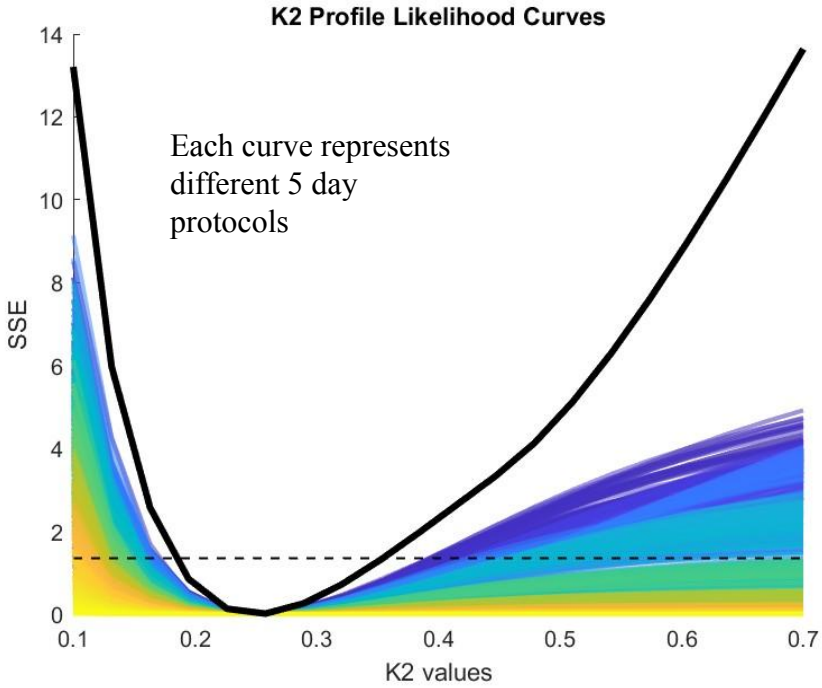
Non-Optimal Experimental Designs: 3 Data Points

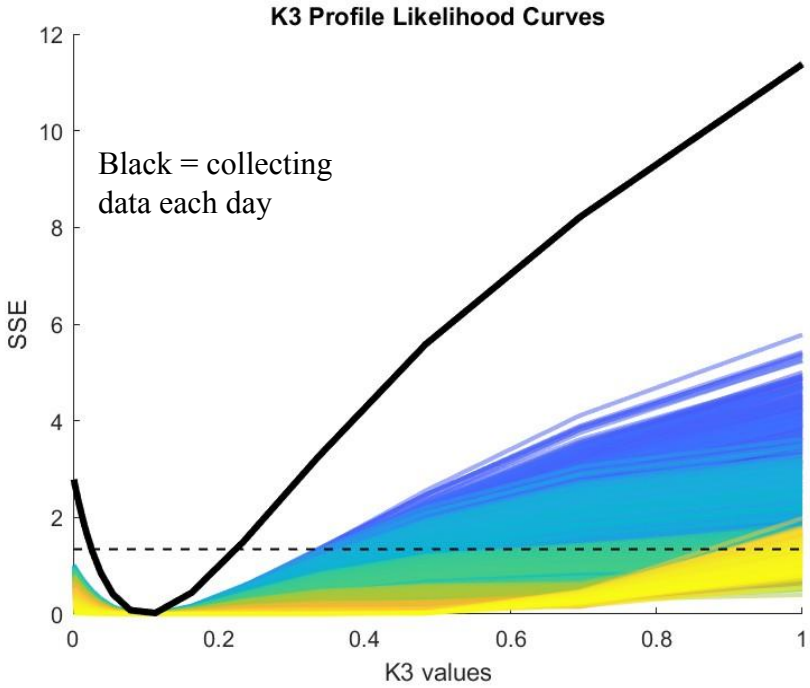




[Note: represents a subset of 3-day protocols.] Although k_2 does exhibit that parabolic shape and crosses threshold for many protocols, we also need both k_3 to be identifiable. The left-end of k_3 does not meet the threshold.

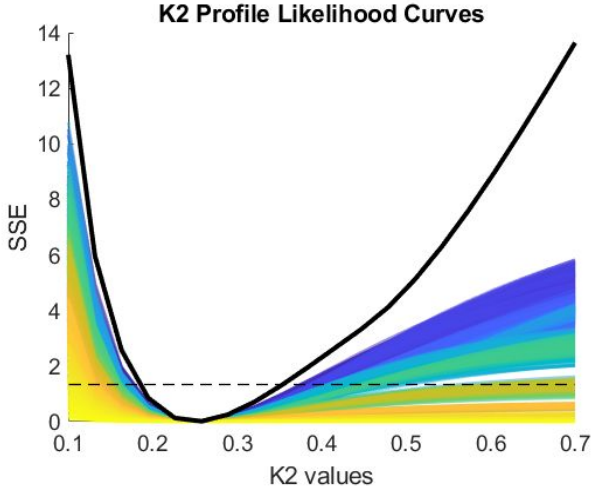
Non-Optimal Experimental Designs: 5 Data Points



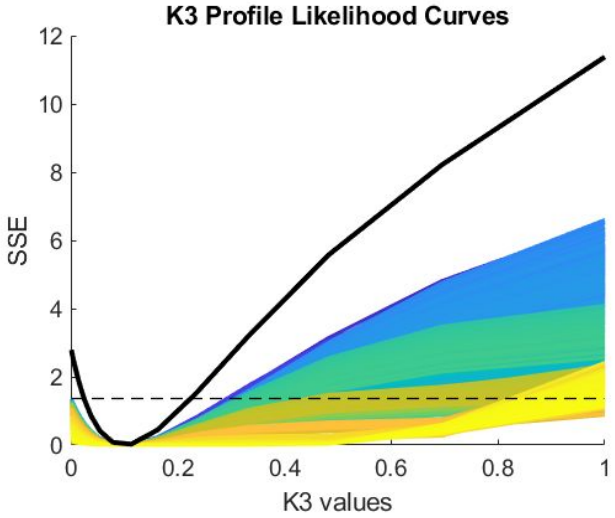


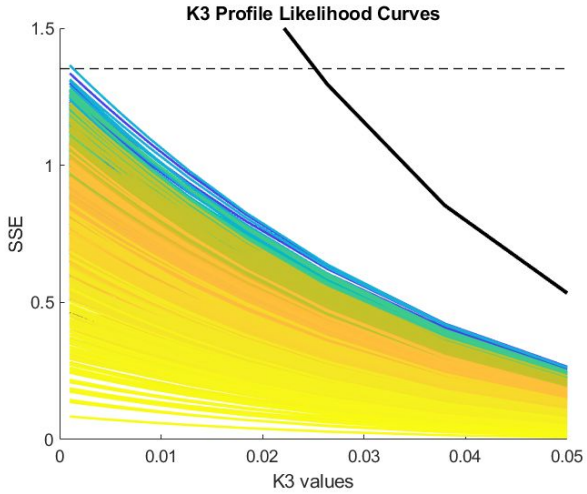
[Note: represents a subset of 5-day protocols.] Still the same problem where the left-end of k_3 does not meet the threshold, so we must collect more data!

Optimal Experimental Designs: 7 Data Points



Each curve represents different 7 day protocols

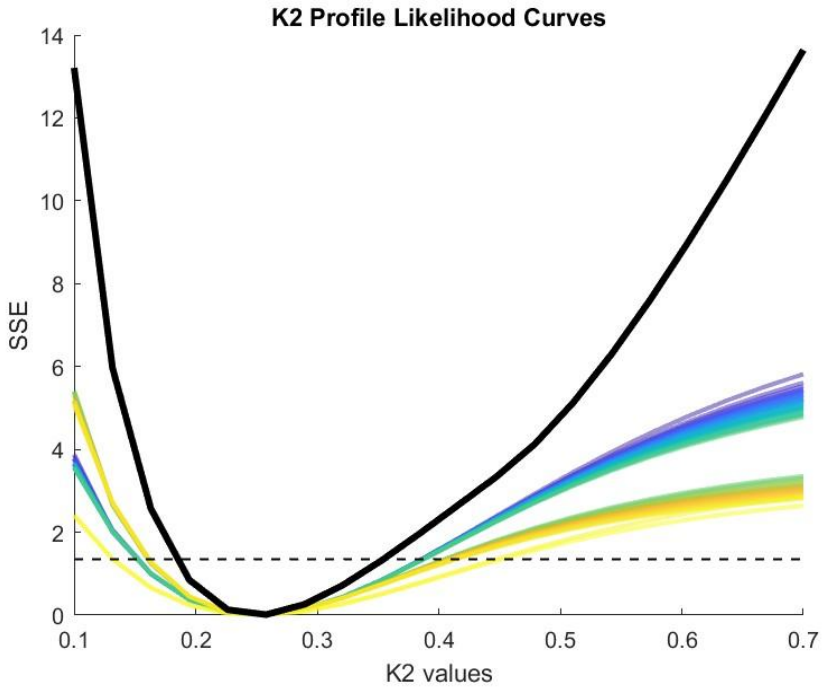


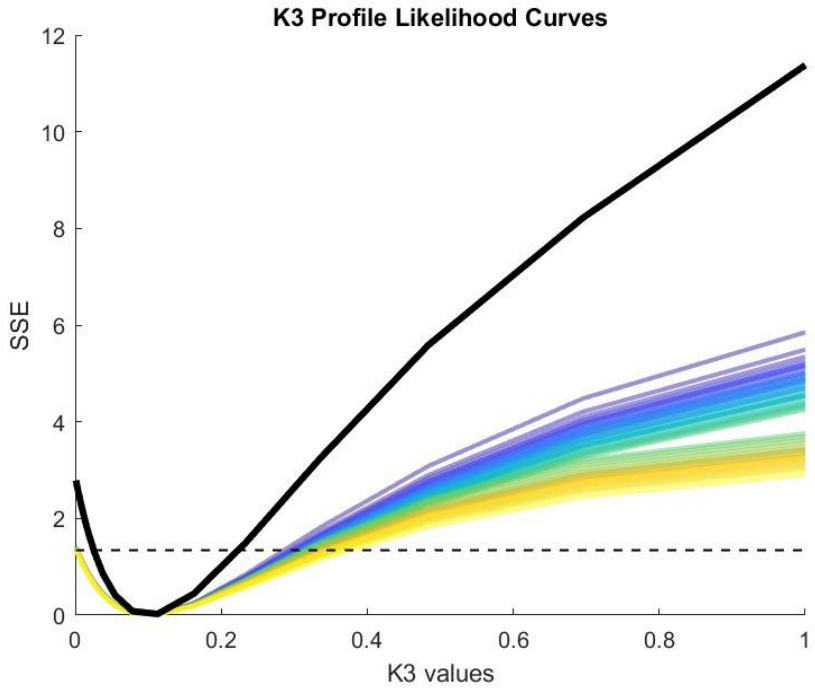


Black = collecting data each day

[Note: represents 2% of the points.] There is at least one where k_3 does meet the threshold.

Profile Likelihood Curve for 7-day Subsets



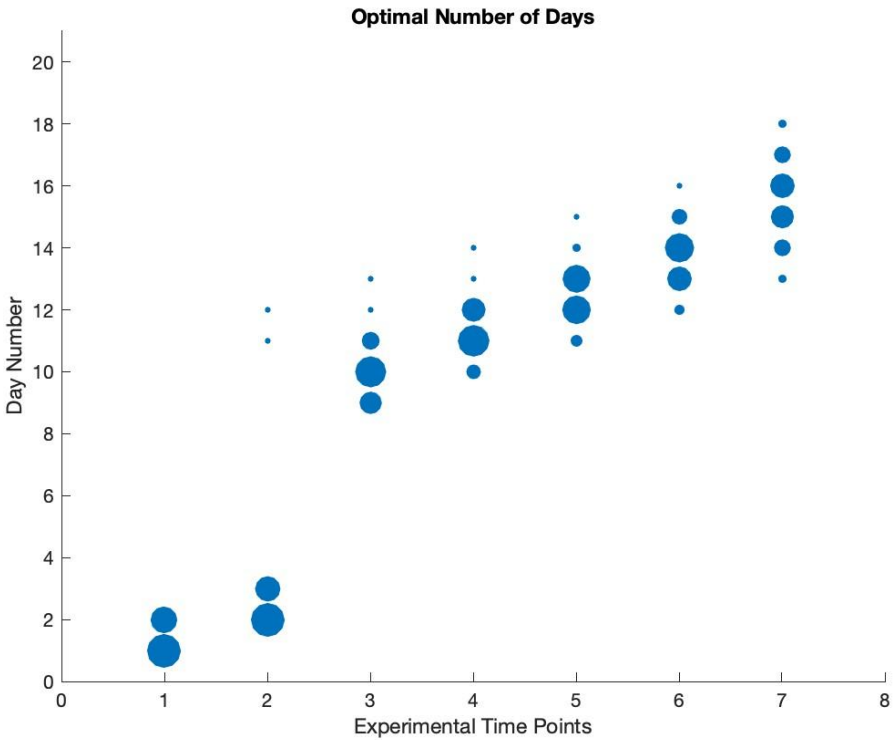


With 7 days of data collection, the left-end of k_3 was able to pass the threshold. Note that among 77,520 7-day combinations, only 52 of them (0.0067%) result in identifiable parameters.

Optimal Experimental Design

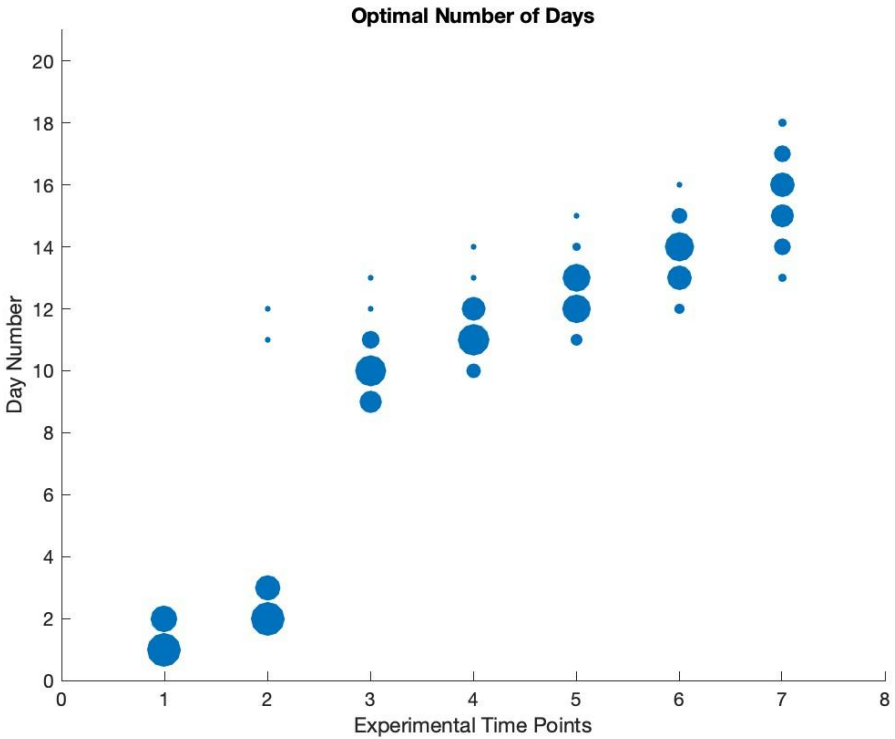
Which 7-day protocols actually result in confident predictions of k_2 and k_3?

- For instance, we find the first measurement (t =1 on horizontal axis) must be taken at either Day 1 or Day 2. If that is not done, the experimental protocol will not result in practically identifiable parameters.
- The second measurement almost universally must be taken the day after the first (though there are a few exceptions).
- Then we take a pretty big jump - not needing to collect data again to a minimum of Day 9.



Conclusions

- Our “minimally sufficient experimental design” analysis results in actionable suggestions for experimentalists.
- For the pharmacokinetic/ pharmacodynamic model we developed, we learn that we need 7 days of measurements of drug concentration in the plasma to trust our model parameterization, and thus trust its predictions.
- We further can suggest which 7 days, among a 20-day window, should be data collection data!



Acknowledgements

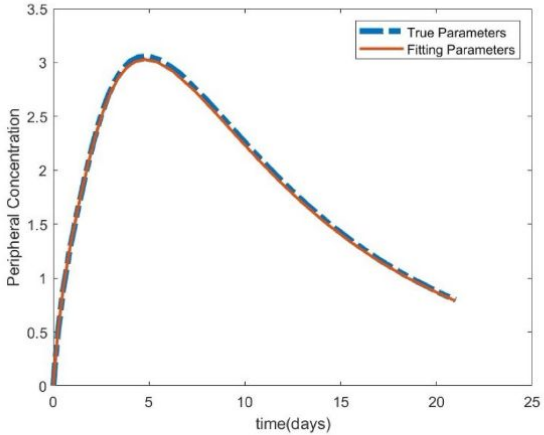
We would like to thank Dr. Gevertz for her guidance on this project for the past 2 semesters!

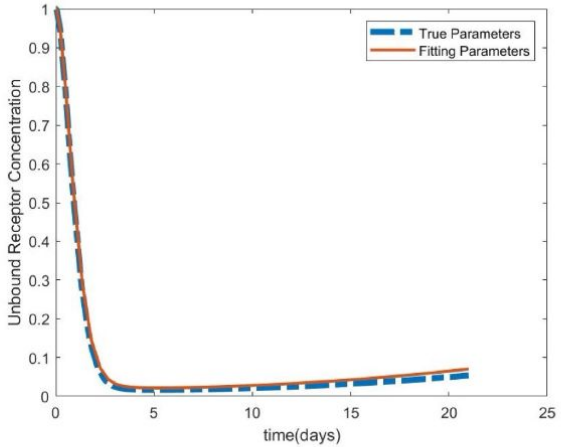
We would like to thank the TCNJ Math Department for providing us with this opportunity and access to the ELSA cluster!

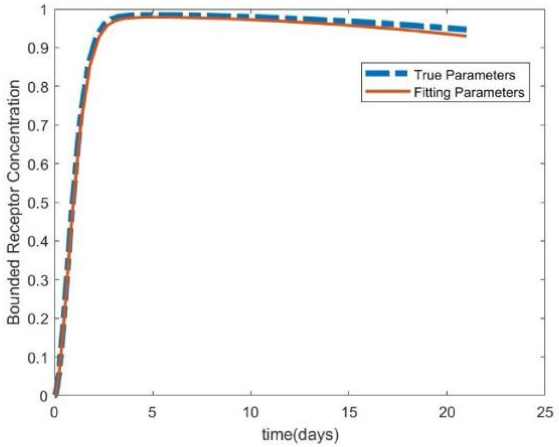


OAC-1826915 and OAC-2320244

Other Slides


True Parameters
Fitting Parameters
Peripheral Concentration
time(days)
0
0.5
1
1.5
2
2.5
3
3.5
0
5
10
15
20
25

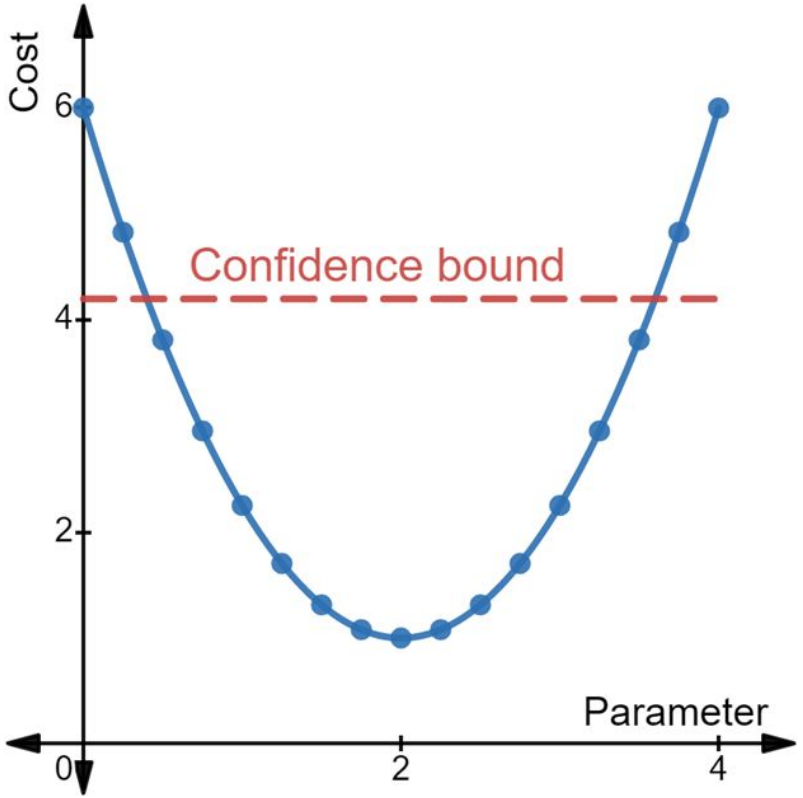

True Parameters
Fitting Parameters
Unbound Receptor Concentration
time(days)
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
0
5
10
15
20
25


True Parameters
Fitting Parameters
Bounded Receptor Concentration
time(days)
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
0
5
10
15
20
25

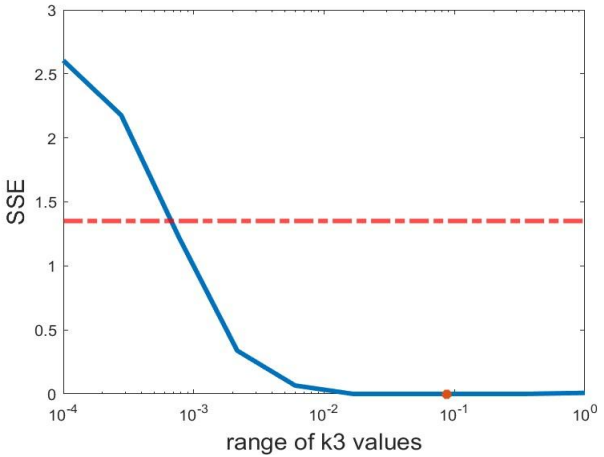
Assessing Practical Identifiability: Profile Likelihood

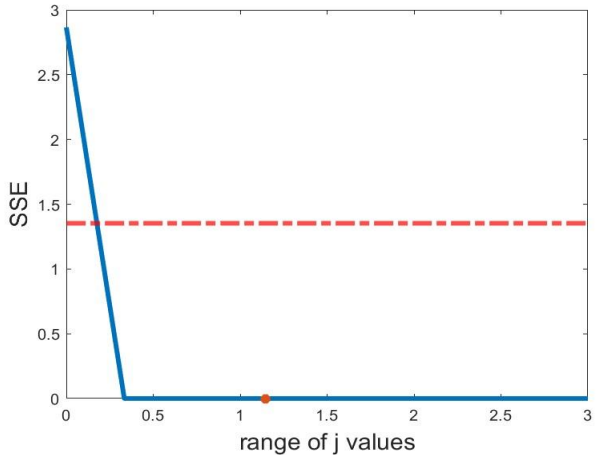
Profile likelihood method for parameter $\boldsymbol{\theta}$

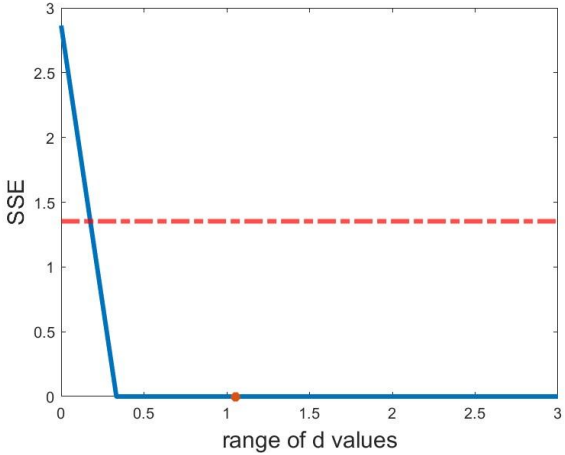
1. Determine a range for the values of $\boldsymbol{\theta}$
2. Fix $\boldsymbol{\theta}=\boldsymbol{\theta}^*$ where $\boldsymbol{\theta}^*$ is a value in the range
3. For the fixed value in Step 2, find the parameter set that minimizes the cost function
4. Save the optimal value of the cost function for $\boldsymbol{\theta}= \boldsymbol{\theta}^*$
5. Repeat the process in Steps 2 – 4 for a discrete set of values in the range of the parameter $\boldsymbol{\theta}$



Identifiability of our Model Parameters







Back to Goal: If we want to replicate the experiment, what is the minimal number of days of data collection needed to preserve identifiability of the subset of parameters?

Narrowing Down to a Subset of Model Parameters

- We've seen that, even in the "ideal" scenario of drug concentration in the central compartment collected daily, we do not have enough data to confidently parameterize the model. Not a single parameter was practically identifiable, and several were not even structurally identifiable.
- Without identifiable parameters, we really cannot trust predictions made by the model in new scenarios, as different model parameterizations that well-describe the data could make very different predictions!
- We turn to biology to help here - there are some parameters in any model that can be well-approximated from experimental data. In our model, the parameters that are easier to measure are kon and koff, the binding and dissociation rates via formulas. We will *fix* these parameters going forward, and then see what happens if we have "ideal" experimental data.